

Assignment 3

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```
bank <- read.csv("C:/Users/brumb/OneDrive/Documents/R/UniversalBank-1.csv")

bank <- bank %>%select(Online, CreditCard, Personal.Loan)

bank <- bank %>%
mutate(Online = factor(Online),CreditCard = factor(CreditCard),Personal.Loan = factor(Personal.Loan))

set.seed(1)
trainIndex <- sample(1:nrow(bank), 0.6*nrow(bank))
train <- bank[trainIndex, ]
valid <- bank[-trainIndex, ]
```

A.Pivot Table

```
tab <- table(train$CreditCard, train$Personal.Loan, train$Online)
ftable(tab)
```

```
##           0      1
##
## 0 0      805 1119
##   1       79  119
## 1 0      332  469
##   1       30   47
```

B.Probability from Pivot Table

```
num_B <- tab["1","1","1"]
den_B <- sum(tab["1", , "1"])
prob_B <- num_B / den_B
prob_B
```

```
## [1] 0.09108527
```

The probability of accepting the loan given CC = 1 and Online = 1 is shown above.

C.Separate Pivot Tables

```
tab_online <- table(train$Personal.Loan, train$Online)
tab_cc <- table(train$Personal.Loan, train$CreditCard)
```

```
tab_online
```

```
##
##      0      1
##  0 1137 1588
##  1  109  166
```

```
tab_cc
```

```
##
##      0      1
##  0 1924  801
##  1  198   77
```

D.Conditional Probabilities

```
pcc1givenloan1 <- tab_cc["1","1"] / sum(tab_cc["1",])
ponline1givenloan1 <- tab_online["1","1"] / sum(tab_online["1",])
ploan1 <- sum(train$Personal.Loan == 1) / nrow(train)
pcc1givenloan0 <- tab_cc["0","1"] / sum(tab_cc["0",])
ponline1givenloan0 <- tab_online["0","1"] / sum(tab_online["0",])
ploan0 <- 1 - ploan1
pcc1givenloan1
```

```
## [1] 0.28
```

```
ponline1givenloan1
```

```
## [1] 0.6036364
```

```
ploan1
```

```
## [1] 0.09166667
```

```
pcc1givenloan0
```

```
## [1] 0.293945
```

```
ponline1givenloan0
```

```
## [1] 0.5827523
```

```
ploan0
```

```
## [1] 0.9083333
```

E.Naive Bayes Calculation

```
num_E <- pcc1givenloan1 * ponline1givenloan1 * ploan1
```

```
den_E <- num_E + (pcc1givenloan0 * ponline1givenloan0 * ploan0)
```

```
prob_E <- num_E / den_E
```

```
prob_E
```

```
## [1] 0.09055758
```

F.Comparison

```
prob_B
```

```
## [1] 0.09108527
```

```
prob_E
```

```
## [1] 0.09055758
```

The pivot table result is more accurate because it uses the true joint distribution. The Naive Bayes result assumes independence between Online and CreditCard given Loan.

G.Naive Bayes Model

```
model <- naiveBayes(Personal.Loan ~ CreditCard + Online, data = train)
```

```
model
```

```
##
```

```
## Naive Bayes Classifier for Discrete Predictors
```

```
##
```

```
## Call:
```

```
## naiveBayes.default(x = X, y = Y, laplace = laplace)
```

```
##
```

```
## A-priori probabilities:
```

```
## Y
```

```
##           0           1
```

```
## 0.90833333 0.09166667
```

```
##
```

```
## Conditional probabilities:
##   CreditCard
## Y          0          1
## 0 0.706055 0.293945
## 1 0.720000 0.280000
##
##   Online
## Y          0          1
## 0 0.4172477 0.5827523
## 1 0.3963636 0.6036364
```

Prediction for CC = 1 and Online = 1

```
prediction <- predict(model,data.frame(CreditCard = factor(1, levels=c(0,1)),Online = factor(1, levels=
prediction
```

```
##          0          1
## [1,] 0.9094424 0.09055758
```

The model predicts only Loan = 0, giving zero sensitivity and failing to identify loan acceptors due to class imbalance, despite appearing accurate overall.

Validation Performance

```
pred_class <- predict(model, valid)

confusionMatrix(pred_class, valid$Personal.Loan, positive="1")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 1795  205
##           1    0    0
##
##           Accuracy : 0.8975
##           95% CI : (0.8834, 0.9105)
##           No Information Rate : 0.8975
##           P-Value [Acc > NIR] : 0.5186
##
##           Kappa : 0
##
## Mcnemar's Test P-Value : <2e-16
##
##           Sensitivity : 0.0000
##           Specificity : 1.0000
##           Pos Pred Value :   NaN
##           Neg Pred Value : 0.8975
##           Prevalence : 0.1025
##           Detection Rate : 0.0000
##           Detection Prevalence : 0.0000
##           Balanced Accuracy : 0.5000
##
```

```
##          'Positive' Class : 1  
##
```

Conclusion

- Only P CC&Loan, P Online&Loan, and P Loan are needed for Naive Bayes
- Pivot table provides the exact probability
- Naive Bayes provides an approximation due to independence assumption
- No transformation (Box-Cox, scaling) is required since all variables are categorical